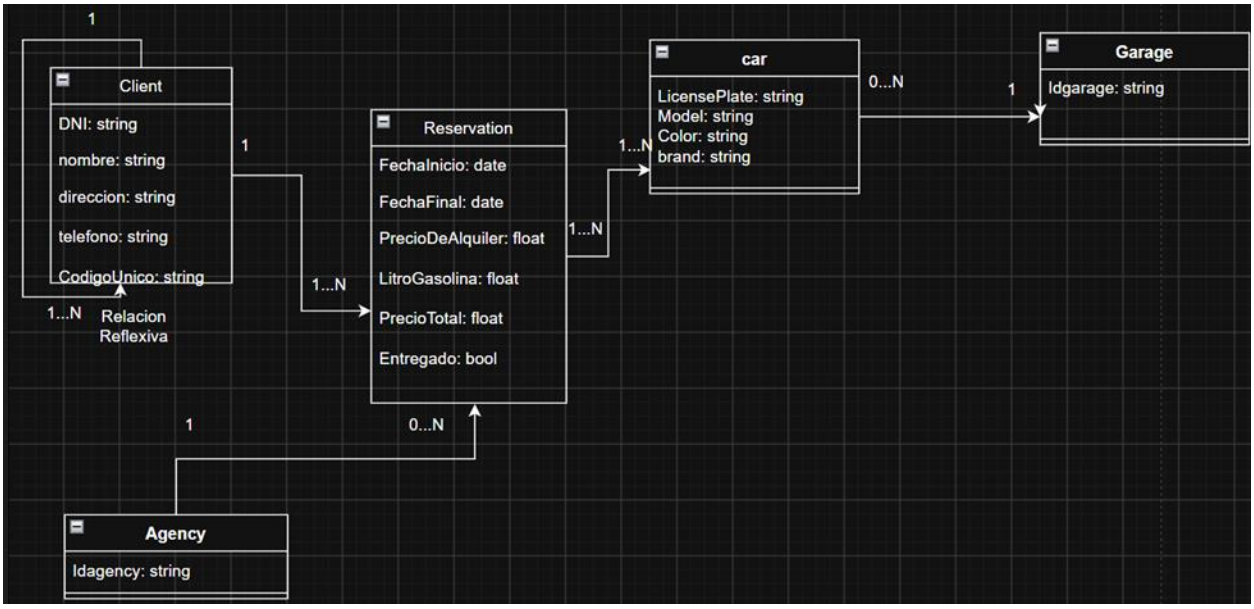
Justification for the Class Diagram

Vehicle Rental Management System



1. Analysis and Justification of Classes and Attributes

Client Class

Purpose: Represents the users or entities that carry out rental transactions within the system.

Attributes and justification:

- **DNI: string: Serves as the official identifier or legal identity document, ensuring the client's legal uniqueness.**
- **nombre: string: Stores the client's full name for identification and contact purposes.**
- **direccion: string: Physical location data required for billing and legal contracts.**
- **telefono: string: Direct communication channel for notifications and reservation management.**
- **CodigoUnico: string: Internal key or member/client number generated by the system to streamline database searches.Clase Reservation (Reserva)**

Purpose: Acts as the central transactional entity linking the customer, the agency, and the selected vehicles.

Attributes and justification:

- **StartDate (date) and EndDate (date): Define the rental period, essential for calculating vehicle availability.**
- **RentalPrice (float): Base rate applied during the rental period.**
- **FuelLiters (float): Tracks fuel level or consumption at the time of pickup or return; useful for additional charge policies.**
- **TotalPrice (float): Final calculated amount incorporating rates, fuel costs, and additional charges.**
- **Delivered (bool): Boolean indicator (flag) showing whether the vehicle has been physically handed over to the customer, facilitating operational management.**

Clase car (Vehículo)

Purpose: Models the inventory of cars available for rent in the fleet.

Attributes and justification:

- **LicensePlate: string: License plate number, used as the vehicle's unique and distinct physical identifier.**
- **Model: string: Specific car model.**
- **Color: string: Vehicle's exterior color.**
- **brand: string: Car manufacturer/brand; essential attributes for the customer to identify and filter for their preferred vehicle.**

Clase Garage (Garaje)

Purpose: Represents the physical infrastructure—such as a storage facility, workshop, or parking area—where vehicles are kept when not in use or undergoing maintenance.

Attributes and justification:

- **Idgarage: string: Unique identifier for the garage or storage facility**

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Clase Agency (Agencia)

- **Purpose: Represents the branch or administrative organization responsible for managing operations, contracts, and personnel.**
- **Attributes and justification:**

Idagency: string: Unique identifier for the agency branch.

2. Justification of Relationships and Multiplicities

Client–Reservation (1:N): A client can make zero or multiple reservations over time (rental history), whereas each reservation belongs mandatorily to a single registered client.• Agency to Reservation (1 to 0...N): An agency manages operational reservations. A specific reservation is processed and administered by a specific agency, whereas an agency may manage multiple reservations—or none—at any given time.

Reservation to Car (1...N to 1...N): A many-to-many relationship justified by the fact that a reservation can include one or more vehicles (e.g., multiple rentals for fleets or groups), and a vehicle can be associated with different reservations at various times throughout its operational lifecycle.

Car to Garage (0...N to 1): A vehicle is assigned to or housed in a specific garage at any given time, while a garage can accommodate anywhere from zero to multiple vehicles, depending on the capacity of its physical infrastructure.

3. conclusión: Two main issues arose regarding this diagram: first, a different logic was involved; second, the option to treat a class as a simple attribute—provided it contains only one attribute, such as the 'Agency' or 'Garage' class—was considered; additionally, the multiplicity settings for some classes could be confusing.